Power

File: power.kicad_sch

MCU

File: mcu.kicad_sch

Driver 1 and matrix

File: matrix1.kicad_sch

Driver 2 and matrix

File: matrix2.kicad_sch

Driver 3 and matrix

File: matrix3.kicad_sch

Parallelogram 4-layer density study (p3-162-usb): 162 XL-1010RGB LEDs (9x18), 3x AW20198QNR, USB-C + Micro-JST 1.25 + NW/SE flat-face wing buttons, 4L.
25 x 70 mm tip-to-tip skewed parallelogram, 20 x 35 mm prismatic LiPo cell.
Solid GND thermal pads on U3, U5, and all AW20198QNR QFN-40 drivers (0 via-in-pad).

Drawn by scripts/parallelogram/sch_gen.py

162 XL-1010RGB LEDs (9x18), 3x AW20198QNR, USB-C + Micro-JST 1.25 + NW/SE flat-face wing b

CERN-OHL-P-2.0

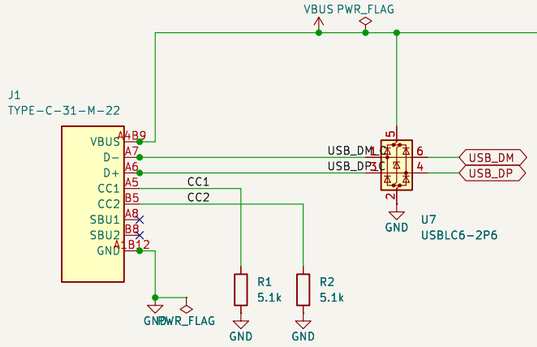
Sheet: /

File: parallelogram.kicad_sch

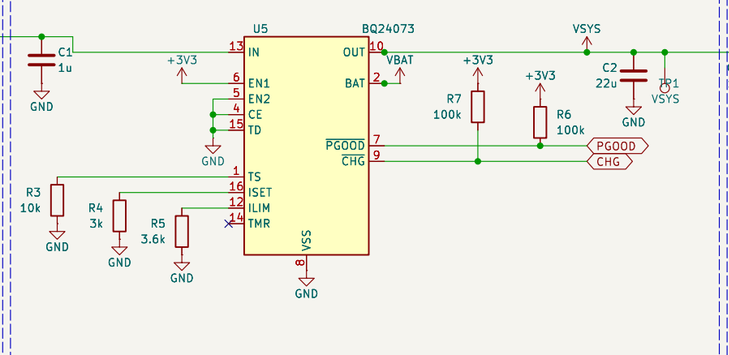
Title: parallelogram (p3-162-usb): top

Size: A4	Date: 2026-11-05	**Rev: P2U**
KiCad E.D.A. 9.0.8		Id: 1/6

USB-C 16P SMT & ESD

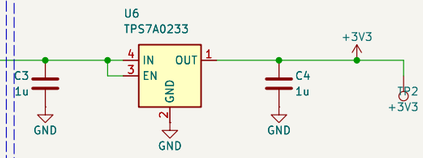


CHARGER AND POWER PATH



USB500: 450–500 mA in, 297 mA charge, ILIM must not float.
TMR open: default 5 h. TS fixed: no NTC.

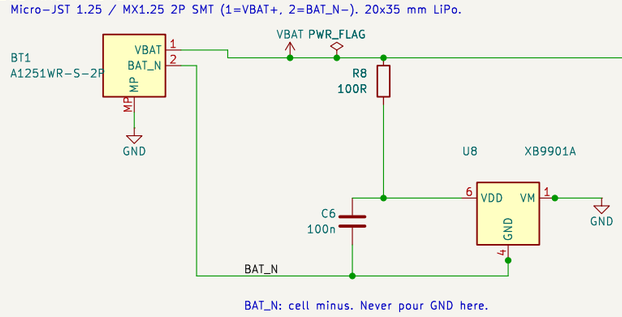
3V3



3V3 follows VSYS down: AW20198 inputs stay within VCC + 0.3 V.
150 mA max.

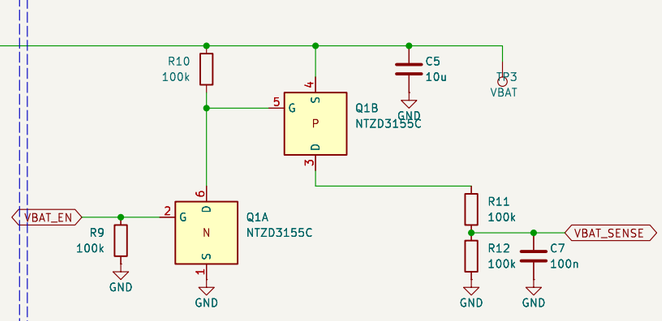
CELL AND PROTECTION

Micro-JST 1.25 / MX1.25 2P SMT (1=VBAT+, 2=BAT_N−). 20x35 mm LiPo.



BAT_N: cell minus. Never pour GND here.

VBAT SENSE



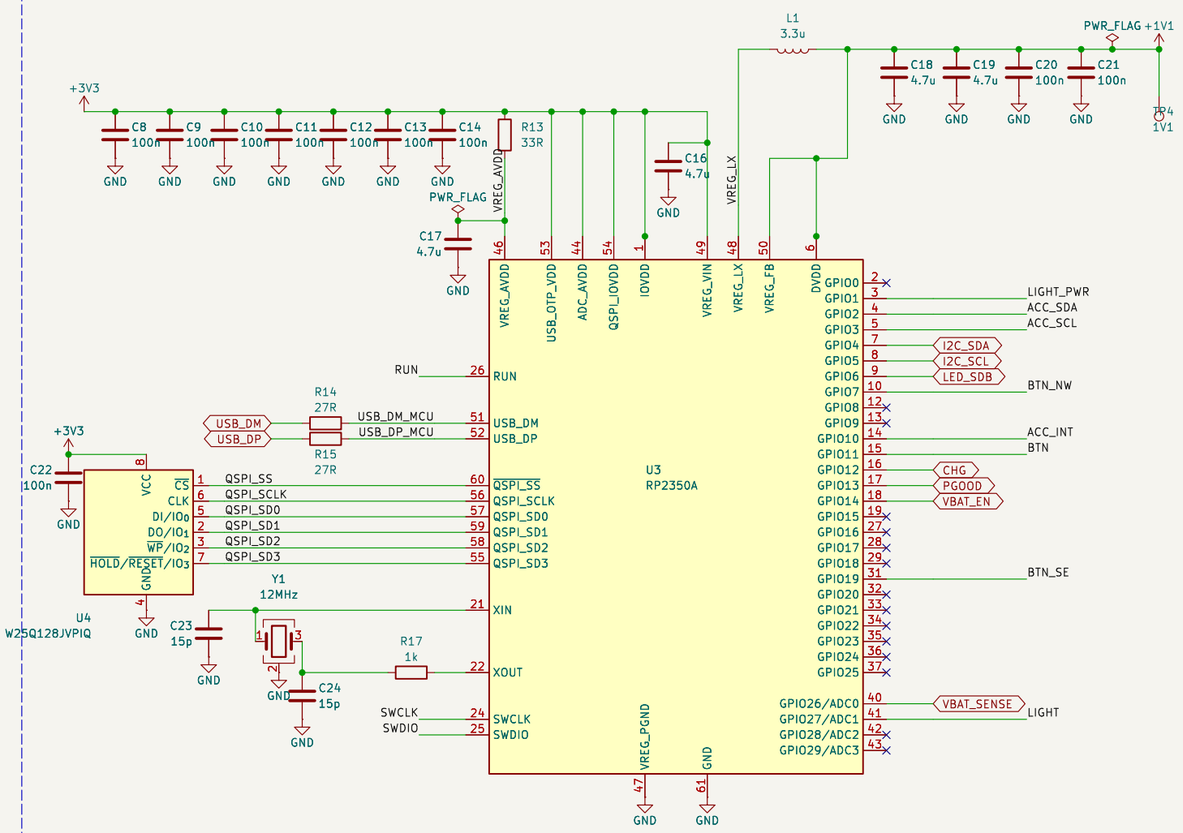
Drawn by scripts/parallelogram/sch_gen.py
USB-C 16P SMT, charger, protection, LDO, Micro-JST 1.25 battery, VBAT sense
CERN-OHL-P-2.0

Sheet: /Power/
File: power.kicad_sch

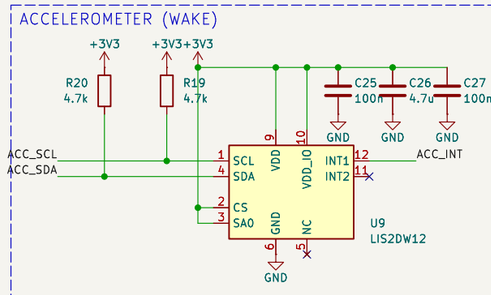
Title: parallelogram (p3-162-usb): power

Size: A3	Date: 2026-11-05	**Rev: P2U**
KiCad E.D.A. 9.0.8		Id: 2/6

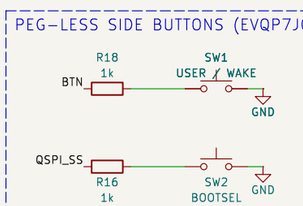
RP2350A, FLASH, CRYSTAL



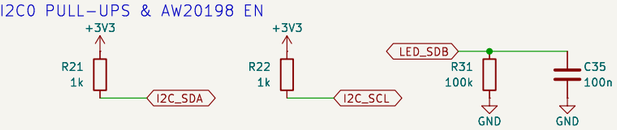
ACCELEROMETER (WAKE)



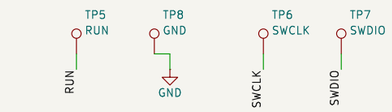
PEG-LESS SIDE BUTTONS (EVQP7J01P)



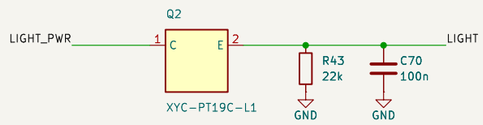
I2C0 PULL-UPS & AW20198 EN



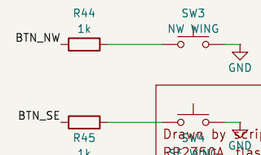
TEST PADS



AMBIENT LIGHT SENSOR (TOP BEZEL)



BACK WING FLAT-FACE BUTTONS (TS-1187A-B-A-B)



Drawn by scripts/parallelogram/sch_gen.py
RP2350A, flash, crystal, accelerometer, peg-less side buttons, top-bezel light sensor

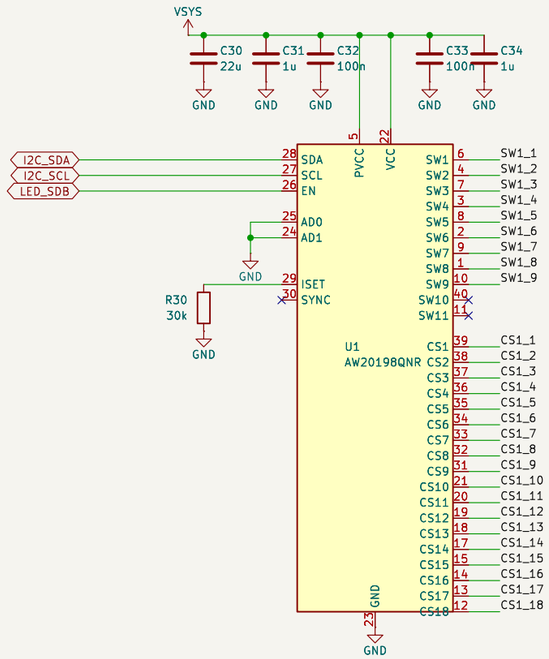
CERN-OHL-P-2.0

Sheet: /MCU/
File: mcu.kicad_sch

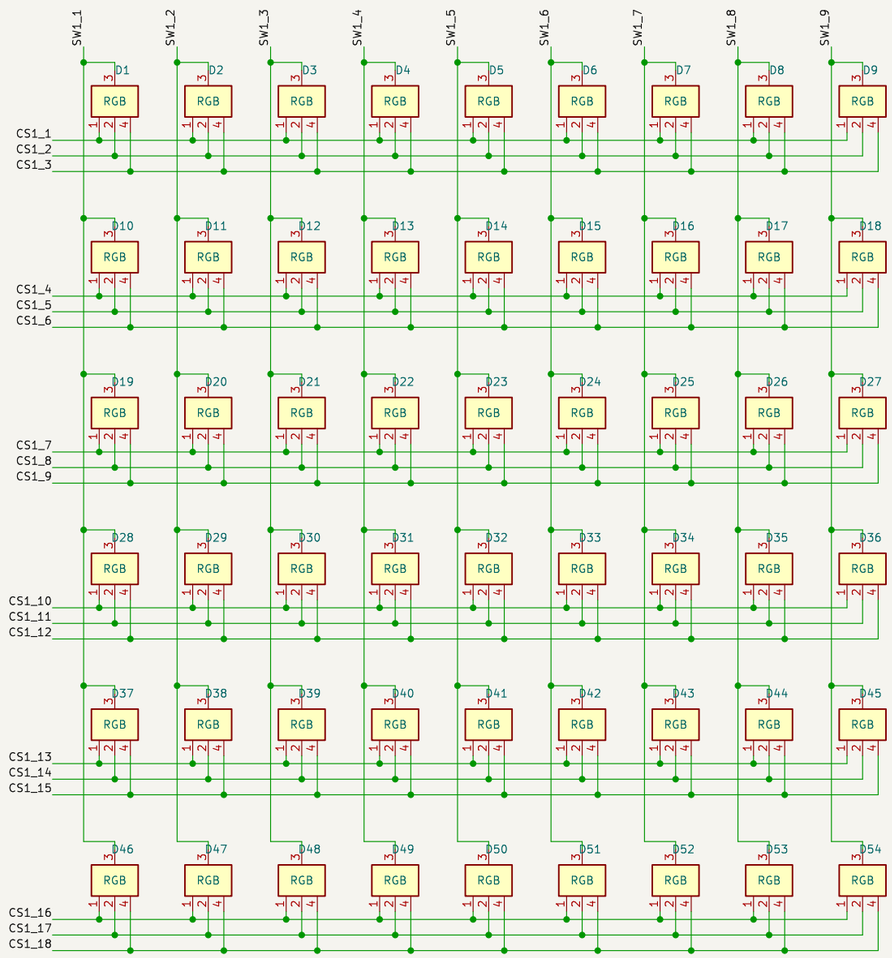
Title: parallelogram (p3-162-usb): mcu

Size: A3	Date: 2026-11-05	**Rev: P2U**
KiCad E.D.A. 9.0.8		Id: 3/6

DRIVER 1 (U1, 0x20)



54 RGB LEDS: R=CS(3k−2), G=CS(3k−1), B=CS(3k)



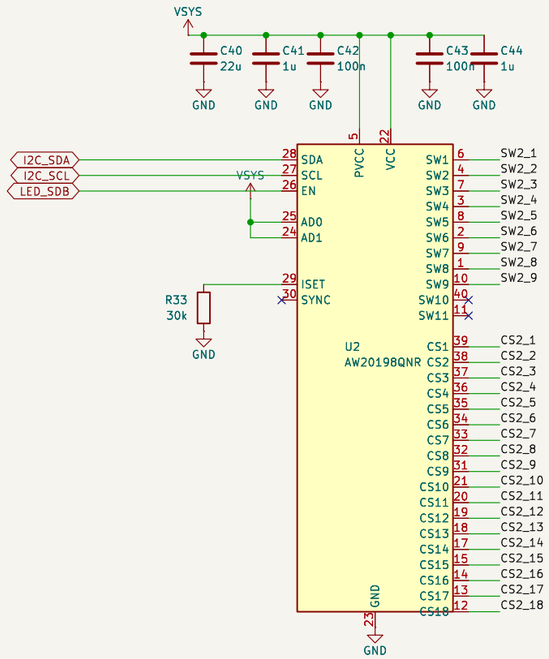
Drawn by scripts/parallelogram/sch_gen.py
U1 AW20198QNR (0x20), 54 RGB LEDs (drivers-aw20198.md)
CERN-OHL-P-2.0

Sheet: /Driver 1 and matrix/
File: matrix1.kicad_sch

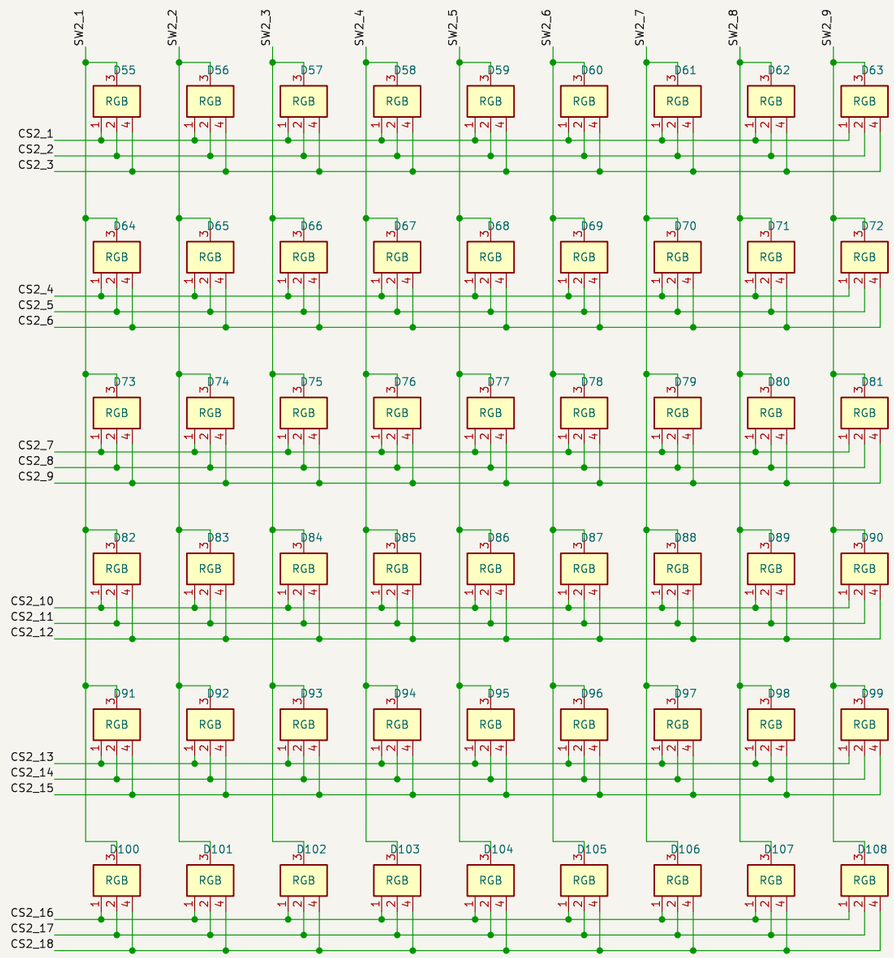
Title: parallelogram (p3-162-usb): driver 1 and matrix

Size: A3	Date: 2026-11-05	**Rev: P2U**
KiCad E.D.A. 9.0.8		Id: 4/6

DRIVER 2 (U2, 0x25)



54 RGB LEDS: R=CS(3k−2), G=CS(3k−1), B=CS(3k)



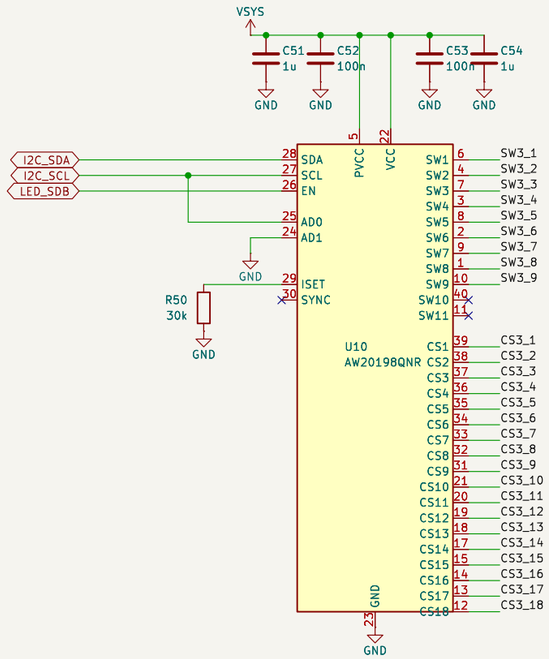
Drawn by scripts/parallelogram/sch_gen.py
U2 AW20198QNR (0x25), 54 RGB LEDs (drivers-aw20198.md)
CERN-OHL-P-2.0

Sheet: /Driver 2 and matrix/
File: matrix2.kicad_sch

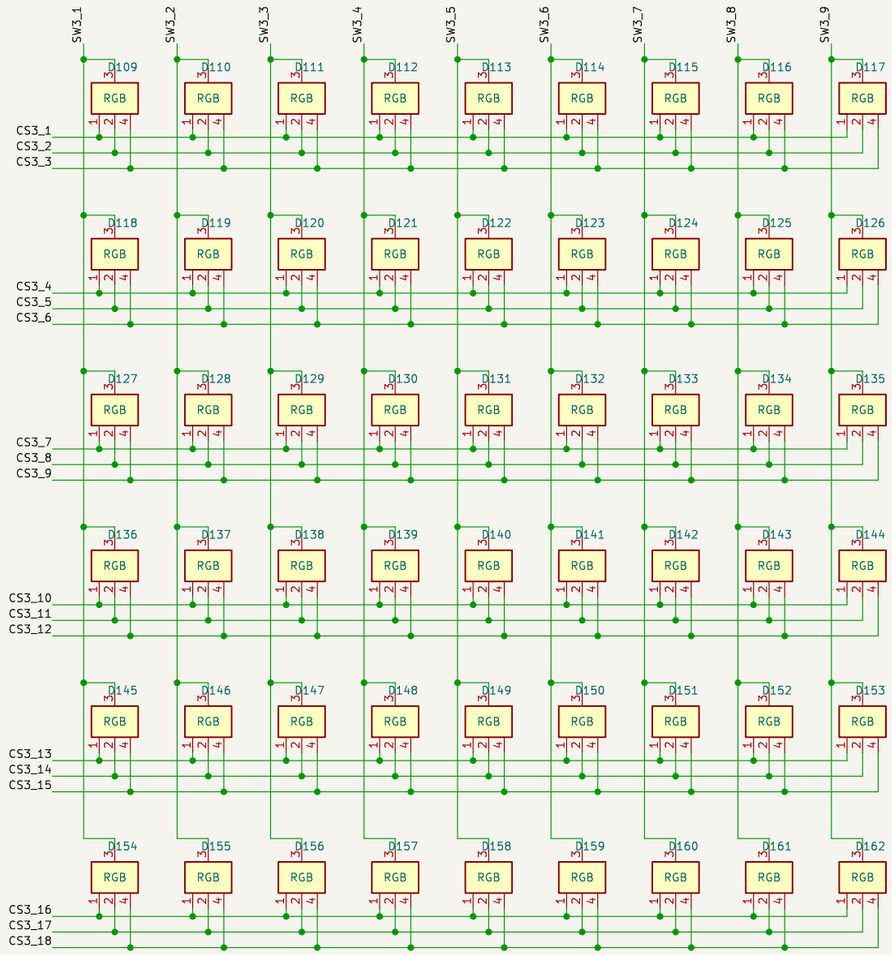
Title: parallelogram (p3-162-usb): driver 2 and matrix

Size: A3	Date: 2026-11-05	**Rev: P2U**
KiCad E.D.A. 9.0.8		Id: 5/6

DRIVER 3 (U10, 0x22)



54 RGB LEDS: R=CS(3k−2), G=CS(3k−1), B=CS(3k)



Drawn by scripts/parallelogram/sch_gen.py
U10 AW20198QNR (0x22), 54 RGB LEDs (drivers−aw20198.md)
CERN−OHL−P−2.0

Sheet: /Driver 3 and matrix/
File: matrix3.kicad_sch

Title: parallelogram (p3−162−usb): driver 3 and matrix

Size: A3	Date: 2026−11−05	**Rev: P2U**
KiCad E.D.A. 9.0.8		Id: 6/6